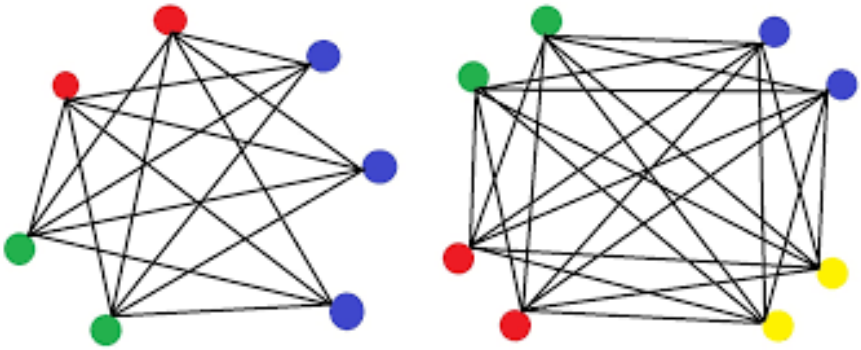
Forbidden Subgraphs in Extremal Graph Theory

Ryan Peng



1 Introduction

Graph theory is an area of combinatorics that studies graphs, which are structures made up of vertices and edges.

Extremal graph theory is a branch of graph theory that studies with how global properties of a graph (such as how many vertices or edges a graph has) are connected with local substructure.

The most fundamental problem in extremal graph theory is the **forbidden subgraph problem**, which asks: How dense can you make a graph before the existence of a specific subgraph is guaranteed? For example, how many edges can a graph have before it must contain a triangle? What about a pentagon? In this article, we will explore these questions.

2 Definitions

2.1 Basics of Graph Theory

To begin, we will define some of the basic terminology of graph theory.

Definition 2.1

A **graph** is an ordered pair $G = (V, E)$, where:

- V is a set of vertices (points)
- E is a set of edges, where an edge is defined as an unordered pair of vertices (each edge connects two distinct vertices).

Definition 2.2

Two graphs G_1 and G_2 are **isomorphic** if there exists a bijection (one-to one mapping) $f : V_1 \to V_2$ between the vertices of the two graphs such that if a,b is an edge in G_1 then $f(a)$, $f(b)$ is an edge in G_2.

Definition 2.3

For an edge x, y, the vertices x and y are said to be the **endpoints** of the edge.

Definition 2.4

Two vertices x and y are **adjacent** if they are connected by an edge.

Definition 2.5

The **degree** of a vertex x (denoted $d(x)$) is the number of edges of which x is an endpoint.

2.2 Named Graphs

Definition 2.6

A graph $G_1 = (V_1, E_1)$ is a **subgraph** of $G_2 = (V_2, E_2)$ if $V_1 \subseteq V_2$ and $E_1 \subseteq E_2$.

Definition 2.7

A **path** (denoted P_n) is a subset of n distinct vertices and $n - 1$ edges, where any two consecutive vertices are adjacent.

Definition 2.8

An n**-cycle** (denoted C_n) can be obtained from a path P_n by connecting the first and last vertex of the path with an edge.

Definition 2.9

An n**-clique** (denoted K_n) is a subset of n vertices where any two vertices in the clique are adjacent.

Definition 2.10

A graph is **bipartite** if it is possible to divide the vertices of the graph into two disjoint sets such that no two vertices in the same set are adjacent.

Definition 2.11

The **complete bipartite graph** $K_{m,n}$ is a bipartite graph whose vertices can be divided into two disjoint sets of size m and n such that every vertex in the first set is adjacent to every vertex in the second set.

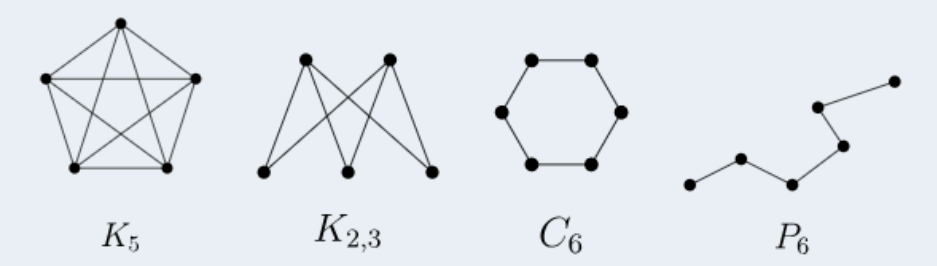


Figure 1: Examples of named subgraphs

2.3 Forbidden Subgraphs

We now define the central focus of this article: the extremal number of a forbidden subgraph.

Definition 2.12

The **extremal number** (denoted $\text{ex}(n, G)$) or **Turán number** (denoted $t(n, G)$) of a graph G is the maximum possible number of edges of a graph on n vertices that does not contain a subgraph isomorphic to G.

3 Results

Now that the definitions are out of the way, we can start proving some results!

3.1 Handshaking Lemma

We will start with a fairly obvious assertion: the sum of the degrees of all of the vertices in a graph is equal to twice the number of edges.

Lemma 1: Handshaking Lemma

For any graph G,

$$\sum_{v \in V(G)} d(v) = 2|E(G)|$$

This makes sense, since each edge is comprised of two vertices, so each edge contributes 2 to the sum of degrees.
This observation is very useful because it allows us to relate the degrees of individual vertices of a graph to the total number of edges in the graph. The idea of relating properties of individual components of a graph to properties of the whole graph is central to extremal graph theory.

(Note: This statement is called the handshaking lemma because of the following analogy:
In a party where people shake hands with each other, if you count the number of handshakes each person received and sum them, the result is twice the total number of handshakes.)

3.2 Mantel's Theorem

For our first forbidden subgraph, we will consider one of the simplest graphs: the triangle. The triangle (or K_3 or C_3) consists of three vertices and 3 edges.

Mantel's theorem was published in 1907 by dutch mathematician Willem Mantel.

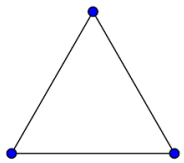


Figure 2: A triangle.

Theorem 1: Mantel's Theorem

The maximum number of edges that a graph on n vertices which does not contain any triangles can have is $\left\lfloor \frac{n^2}{4} \right\rfloor$. In other words, $\text{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$.

Proof: Let G be a triangle-free graph on n vertices. Since we are looking to prove an upper bound on the number of edges G can have, assume G has at least one edge.

First, consider an arbitrary edge of G. Call the two vertices that make up this edge x and y.
Now consider a third vertex z. If z is adjacent to both x and y, then x, y, z would form a triangle!

Therefore, no vertex of G can be adjacent to both x and y. In other words, the total number of vertices adjacent to x plus the total number of vertices adjacent to y cannot exceed the total number of vertices in G.
This leads us to our first key observation:

$$d(x) + d(y) \leq n$$

Now, how can we relate this to the total number of edges in G? In order to use the handshaking lemma, we need the sum of degrees over all vertices of G, not just two vertices.

So, let's consider taking the sum of $d(x) + d(y)$ over all edges x,y in G. Call this sum S.

$$S = \sum_{\{x,y\} \in E(G)} d(x) + d(y)$$

Since each $d(x) + d(y)$ is less than or equal to n, we have $S \leq n|E(G)|$.

Now, let's look at this sum another way. Consider an arbitrary vertex z of G. Since there are $d(z)$ edges containing z, the term $d(z)$ will be added to S $d(z)$ times. Therefore,

$$S = \sum_{z \in V(G)} d(z)^2$$

Substituting this into our inequality, we have

$$\sum_{z \in V(G)} d(z)^2 \leq n|E(G)|.$$

(Note: this technique of expressing the same sum in two different ways and then combining the two expressions is called **double counting**. It is a useful proof technique in combinatorics.)

Now, we just need to get rid of the power of 2, and then we can use the handshaking lemma. To do this, we can use the Cauchy-Schwarz inequality:

Lemma 2: Cauchy-Schwarz Inequality

Let $x_1, x_2, ..., x_n$, $y_1, y_2, ..., y_n$ be real numbers. Then

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right).$$

Setting $x_1, x_2, ..., x_n$ as the degrees of the vertices in G and $y_1, y_2, ..., y_n = 1$, we obtain

$$n \cdot \sum_{z \in V(G)} d(z)^2 \geq \left(\sum_{z \in V(G)} d(z)\right)^2$$

Therefore

$$\left(\sum_{z \in V(G)} d(z)\right)^2 \leq n^2 |E(G)|$$

To finish, we apply the handshaking lemma:

$$(2|E(G)|)^2 \leq n^2 |E(G)|$$

$$4|E(G)|^2 \leq n^2 |E(G)|$$

$$4|E(G)| \leq n^2$$

$$|E(G)| \leq \frac{n^2}{4}$$

□

The graph that maximizes the number of edges without containing any triangles is the complete bipartite graph $K_{\frac{n}{2},\frac{n}{2}}$.

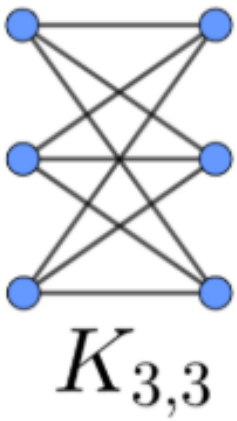


Figure 3: A triangle-free graph on 6 vertices with $\frac{6^2}{4} = 9$ edges.

In this graph, each vertex has degree $\frac{n}{2}$, so the total number of edges is $\frac{1}{2}(n \cdot \frac{n}{2}) = \frac{n^2}{4}$. The floor function is only necessary when n is odd, in which case the number of edges is maximized in $K_{\frac{n+1}{2},\frac{n-1}{2}}$.

3.3 Turán's Theorem

Mantel's theorem was generalized for cliques of all sizes by Hungarian mathematician Pál Turán in 1941.

Theorem 2: Turán's Theorem

$$\mathrm{ex}(n, K_r) = \left(1 - \frac{1}{r-1} + o(1)\right)\left(\frac{n^2}{2}\right).$$

$o(1)$ (pronounced "little-o of 1") denotes a function that approaches 0 as n approaches infinity. Adding the term $o(1)$ accounts for error when n is not divisible by $r-1$, similar to the use of the floor function in Mantel's theorem. This error term does not affect the proof, so we will ignore it.

Proof: We will prove the theorem by showing that for any graph G on n vertices, $|E(G)|$ is less than or equal to our claimed value of $\mathrm{ex}(n, G)$. We will show this by strong induction on n.

For the base case, consider a graph G with n vertices, where $n < r$.

A graph with fewer than r vertices cannot possibly contain K_r, so the maximum number of edges is simply $\binom{n}{2}$. Therefore we have

$$|E(G)| \leq \binom{n}{2}$$

$$|E(G)| \leq \frac{n(n-1)}{2}$$

$$|E(G)| \leq \left(1 - \frac{1}{n}\right)\left(\frac{n^2}{2}\right)$$

Since $n < r$, $n \leq r-1$, so $\frac{1}{n} \geq \frac{1}{r-1}$. Therefore

$$|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\left(\frac{n^2}{2}\right).$$

Now for our induction hypothesis, assume that the theorem holds for all $n \leq k$ for some positive integer k. We want to show that the theorem then holds for $n = k+1$.

Consider a graph G on $k+1$ vertices that does not contain K_r. We want to show that

$$|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}.$$

Suppose G has a maximal number of edges. Then G must contain K_{r-1}, or else we would be able to add more edges.
Now, split the vertices in G into two sets A and B, where A contains $r-1$ vertices that form a K_{r-1} and B contains the remaining $k-r+2$ vertices in G.

Then the edges in G can be split into three categories: edges in A, edges in B, and edges between A and B. We will consider each of these categories separately, and then take the sum to get the total number of edges.

Since A is a clique, the number of edges in A is simply

$$\binom{r-1}{2} = \frac{(r-1)(r-2)}{2}.$$

Since B contains no K_r, we can apply the inductive hypothesis. Therefore, the number of edges in B is at most

$$\left(1 - \frac{1}{r-1}\right)\frac{(k-r+2)^2}{2}$$

To find the number of edges between A and B, consider an arbitrary vertex x in B. If x was adjacent to every vertex in A, then $x \cup A$ would be a clique of size r, which is not allowed. Therefore, every vertex in B is adjacent to at most $r-2$ vertices in A, so the number of edges between A and B is at most

$$(k-r+2)(r-2).$$

Adding these three expressions, we get that the number of edges in G is at most

$$\frac{(r-1)(r-2)}{2} + \left(1 - \frac{1}{r-1}\right)\frac{(k-r+2)^2}{2} + (k-r+2)(r-2)$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)((r-1)^2 + (k-r+2)^2 + 2(r-1)(k-r+2))$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)((r-1) + (k-r+2))^2$$

$$= \frac{1}{2}\left(1 - \frac{1}{r-1}\right)(k+1)^2$$

$$= \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}.$$

Thus, $|E(G)| \leq \left(1 - \frac{1}{r-1}\right)\frac{(k+1)^2}{2}$, completing the induction. □

The graph that maximizes the number of edges without containing K_r is the **Turán graph** $T_{n,r-1}$, a complete $(r-1)$-partite graph on n vertices.

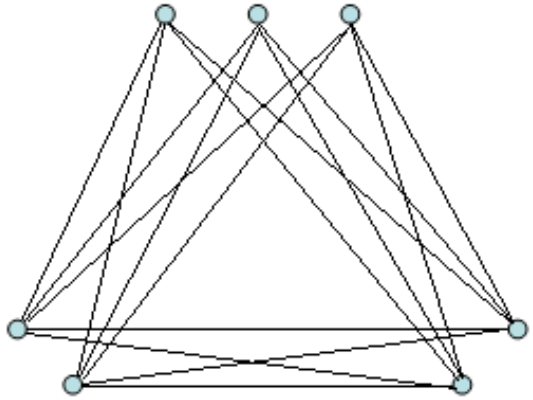


Figure 4: $T_{7,3}$

When n is divisible by $r-1$, each vertex in this graph has degree $n \cdot \left(1 - \frac{1}{r-1}\right)$, so the total number of edges is $\left(1 - \frac{1}{r-1}\right)\frac{n^2}{2}$, the exact value of our upper bound.

(Note: The two diagrams on the first page are of $T_{7,3}$ and $T_{8,4}$.)

3.4 Further Generalization

Now that we've found the extremal number for all cliques, the natural question to ask is: can we generalize this to find the extremal number for all graphs? Well, it turns out there is a theorem that does this, often called the fundamental theorem of extremal graph theory:

Theorem 3: Erdös–Stone–Simonovits Theorem

For any graph G, we have

$$\mathrm{ex}(n, G) = \left(1 - \frac{1}{\chi(G) - 1} + o(1)\right)\left(\frac{n^2}{2}\right)$$

(The proof of this theorem is significantly more involved than the proof of Turán's theorem, so we will not reproduce it in this article.)

To understand this theorem, we need to define $\chi(G)$:

Definition 3.1

The **chromatic number** of a graph G (denoted $\chi(G)$) is the minimum number of unique colors needed to color every vertex of G in such a way that no two adjacent vertices are the same color.

While this definition might seem random at first, it turns out that the chromatic number is the fundamental property of a graph that determines its extremal number!

For example, in a clique, any two vertices are adjacent, so no two vertices can be the same color. Therefore, $\chi(K_r) = r$. Then the Erdös–Stone–Simonovits theorem states

$$\mathrm{ex}(n, K_r) = \left(1 - \frac{1}{r - 1} + o(1)\right)\left(\frac{n^2}{2}\right)$$

This recovers Turán's theorem.

The Erdös–Stone–Simonovits theorem does not tell you the structure of the extremal graph that maximizes the number of edges. In general, this is an open problem.

4 Problems

By constructing graphs to represent pairwise interactions between objects, we can apply results from graph theory to problems that do not explicitly involve graphs. This is a useful technique for solving problems.

For example, consider the following problem:

Problem 1

Let S be a set of n irrational numbers. Find the maximum possible number of unordered pairs (a, b) such that $a, b \in S$ and $a + b$ is rational.

Solution: We can construct a graph G where each number in S corresponds to a vertex in $V(G)$, and two vertices of G are adjacent if and only if their corresponding numbers in S sum to a rational number. Then the problem becomes finding the maximum number of edges that G can have.

Claim: G is triangle-free.

Proof: Assume there exist three irrational numbers x, y, z such that $x+y, y+z, x+z$ are all rational. Since the sum of two rational numbers is also rational, then $(x+y)+(y+z)+(x+z) = 2(x+y+z)$ must be rational, so $x+y+z$ is rational.
Since the difference of two rational numbers must be rational, $(x+y+z)-(x+y), (x+y+z)-(y+z), (x+y+z)-(x+z)$ must all be rational, so x, y, z must all be rational, which is a contradiction.
Therefore there cannot exist three irrational numbers whose pairwise sums are all rational.

Since we know G is triangle-free, we can apply Mantel's theorem, so the maximum number of edges in G is $\boxed{\left\lfloor \frac{n^2}{4} \right\rfloor}$.

Here's another example, this time utilizing Turán's theorem:

Problem 2

In a volleyball tournament with 99 teams, each team plays against each other team exactly once. Every game results in either a win for one of the two teams, or a draw.

Show that if there does not exist a team that draws against more than 8 other teams, then there must exist a set of 11 teams such that no two teams in the set draw against each other.

Solution: We can construct a graph G where each vertex corresponds to a volleyball team, and two vertices are adjacent if their corresponding teams did not draw against each other.
Then we want to prove that G must contain a K_{11}.

Since each team plays 98 other teams, and each team draws against at most 8 other teams, $d(v) \geq 90$ for all vertices $v \in V(G)$.

Then by the handshaking lemma, $\sum_{v \in V(G)} d(v) = 2|E(G)| \geq (99)(90)$, so

$$|E(G)| \geq \frac{(99)(90)}{2}.$$

By Turán's theorem, the maximum number of vertices that a graph on 99 vertices that does not contain K_{11} can have is $\frac{99^2}{2} \cdot \frac{9}{10}$.

For the sake of contradiction, assume that G does not contain K_{11}. Then

$$|E(G)| \leq \frac{99^2}{2} \cdot \frac{9}{10} < \frac{99^2}{2} \cdot \frac{10}{11} = \frac{(99)(90)}{2}.$$

This is a contradiction, so G must contain K_{11}, and we are done.

Lastly, here are some more problems to try on your own:

Problem 3

Let $v_1, v_2, ..., v_n$ be complex numbers such that $|v_i| \geq 1$ for all i, where $|z|$ denotes the magnitude of a complex number z. Prove that there are at most $\lfloor \frac{n^2}{4} \rfloor$ pairs v_i, v_j such that $|v_i + v_j| < 1$.

Problem 4

A graph has 15 vertices. Each edge is colored either red or blue so that there no monochromatic triangles. Find the largest possible number of edges of this graph.

Problem 5

Let n be a positive integer. For a set S of $2n$ real numbers, find the maximum possible number of pairwise (positive) differences between two elements in S, that are in the range $(1, 2)$.

References

[1] Vascimini, Vincent. 2017. Extremal Graph Theory: Turán's Theorem. In *BSU Honors Program Theses and Projects*. Item 234. Available at: https://vc.bridgew.edu/honors_proj/234

[2] Wigderson, Yuval. 2025. Extremal graph theory and Ramsey theory. https://www.ias.edu/sites /default-/files/uss.pdf

[3] Levin, Oscar. 2018. Discrete Mathematics: An Open Introduction. https://discrete. openmath-books.org/dmoi2/ch_graphtheory.html

[4] Quayle, Gridley. 2015. Graph Theory Contest Maths. https://math.stackexchange.com/questions /1101827/graph-theory-contest-maths

[5] Tang, Adrian. 2008. IMO Training 2008: Graph Theory. https://web.mit.edu/yufeiz/www/ imo2008/tang-graph.pdf